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Quiz for Week 4

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1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

1 / 1 point

Let $f(x) = x_1^2 + 2x_2^2$. Find the gradient of $f(x)$ at $(x_1, x_2) = (2, 1)$. Write down a pair of parentheses to include two values separated by a comma. Do not have any symbol other than numeric values, negation, decimal point, comma, and parentheses in your answer. Do not have unnecessary zeros at the end of a number. For example, if you believe the gradient is $(0.1, -2.3)$, write down "$0.1, -2.3$".

4,4

Correct

2. Let's solve

1 / 1 point

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + 2x_2^2$$

by gradient descent. Let $x^0 = (1, 0)$ be the initial solution. Run on iteration of gradient descent to find the next solution x^1. In each iteration, let the step size be that bringing you to the global minimum along the improving direction. Follow the output format requirement provided in Question 1.

0,0

Correct

3. Let $f(x) = x_1^2 + 2x_2^3$. Find the Hessian of $f(x)$ at $(x_1, x_2) = (0, 2)$. Then find its determinant. Write down a single number. Do not have any symbol other than numeric values, negation, and decimal point in your answer. Do not have unnecessary zeros at the end of a number. For example, if you believe the determinant is -9.01, write down "-9.01".

1 / 1 point

48

Correct

4. Let's solve

1 / 1 point

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + 2x_2^3$$

by Newton's method. Let $x^0 = (6, 6)$ be the initial solution. Run on iteration of Newton's method to find the next solution x^1. Follow the output format requirement provided in Question 1.

0,3

Correct

5. Which of the follow statements are correct? Check all correct answers.

1 / 1 point

- [x] Gradient descent is an *interior-point method*, i.e., its search path is not restricted in the boundary of the feasible region of an optimization problem.

Correct

- [ ] Gradient descent is always slower than the Newton's method in solving a nonlinear optimization problem.
- [x] Gradient descent is a first-order method, i.e., it does not require any information about the second-order derivative of the objective function.

Correct

- [x] The Newton's method always find an optimal solution when it is used to solve an unconstrained minimization problem whose objective function is quadratic with upward curvature.

Correct

- [ ] All of the above.